Shun-Hsiang Hsu

Machine Learning Research • Multimodal Learning • Computer Vision
hsus2@illinois.edu | 217-979-3580 | huhuman.github.io

Education

University of Illinois Urbana-Champaign (UIUC) **Aug. 2022 – Present**
Ph.D. in Construction Engineering and Management, GPA: 4.0 **Expected 2027**

- Research in multimodal AI, computer vision, 3D vision, and geometry-grounded synthetic data generation
- Teaching Assistant, Computer Vision (Spring 2024); Government Fellowship for Studying Abroad (2024–2026).

Master of Computer Science, GPA: 4.0 **Aug. 2023 – Dec. 2024**

Professional Experience

Google *Software Engineering Intern (PhD)* Sunnyvale, CA | **Summer 2026**

- Designed controlled experiments for an LLM autorater of hypotheses generated by a troubleshooting agent, comparing scoring formulations and rubric-grounded prompting strategies for repeated-run score stability and alignment with human judgments.
- Iteratively refined evaluation prompts and rubrics through disagreement analysis, achieving over 90% score consistency across repeated evaluations and over 90% alignment with human ratings.
- Investigated scaling bottlenecks in skill-based agents and designed a graph representation that encodes sequencing, dependencies, and other skill relationships beyond prose-only descriptions, enabling relationship-aware context retrieval, orchestration, and automated skill-library health checks.

Reconstruct Inc. *Computer Vision Engineer Intern* Champaign, IL (Hybrid) | **Jan. 2024 – May 2026**

- Evaluated and adapted emerging computer-vision and 3D methods through staged, product-specific experiments to determine technical feasibility before downstream integration.
- Built controlled benchmarks for production image matching and 3D reconstruction, comparing learned feature descriptors and matching algorithms under low-texture, repeated-pattern, sparse-view, and illumination-shift conditions to guide production method selection.
- Integrated FiLM-conditioned camera-orientation cues into a Swin Transformer segmentation model to reduce top- and bottom-view confusion and extended the model to inspection-specific object classes.
- Trained domain-specific visual recognition models for infrastructure and facility inspection using the company's internal active-learning pipeline to iteratively collect labels, retrain models, and expand coverage across object and defect categories.

Selected Research

Geometry-Grounded Synthetic Data for Visual Recognition

- Developed a parameterized 3D synthetic-data framework that generates rendered imagery together with geometry-derived supervision from controllable defect models, including pixel-level masks and physical defect attributes without manual annotation.
- Generated a 72K-image synthetic crack dataset and evaluated synthetic-to-real transfer across three public real-image segmentation benchmarks ranging from hundreds to a few thousand labeled images; realism enhancement yielded zero-shot IoU gains of 3.7–13.6 points.
- Developing a CAD-referenced asset-recognition extension that uses product CAD models to generate multi-view appearance and geometry representations for matching and few-shot recognition, providing structured geometric supervision beyond image-only synthetic data.

Vision-Language Learning for Visual Reporting and Reasoning

- Developed VisualSiteDiary, a detector-free vision-language Transformer for generating textual construction reports directly from visual observations.
- Developed VL-Con, a vision-language dataset and few-shot prompting framework that conditions vision-language model outputs on in-domain visual-text demonstrations for construction monitoring tasks.
- Evaluated single- and multi-agent vision-language workflows for image-to-text inspection reporting, including model-generated self-assessment for identifying low-quality outputs before downstream use.

Additional Experience

TryTech Co., Ltd. *Founder / Technical Lead* Taipei, Taiwan | **Jan. 2021 – Present**

- Built the LLM backend for an omnichannel customer-support prototype, integrating retrieval, tool orchestration, backend transaction workflows, and agent-console handoff for human escalation.
- Developed agent-assisted analysis workflows to surface problematic customer conversations and user feedback, helping identify recurring product and service issues and translate them into targeted fixes and workflow updates.
- Built a customer-service performance analytics pipeline by extracting embeddings from agent responses and training a classifier to identify effective service utterances, enabling automated counts of qualifying interactions for human performance evaluation.
- Designed and operationalized human-AI workflows with domain specialists across creative production and QA, including AI-assisted product reskinning and ticket-based QA validation, and advised teams on integrating model outputs into existing review processes.
- Prototyped AI-enabled applications spanning structural inspection and post-disaster assessment, implementing iOS and web interfaces and deploying machine-learning and custom algorithm modules on AWS using SAM and Lambda for industry pilots.

NCREE–NTUCE Joint AI Research Center *Research Assistant* Taipei, Taiwan | **May 2020 – Jul. 2022**

- Led six industry research projects developing and deploying deep-learning inspection systems for multiple defect types across concrete and steel structures, integrating visual recognition models into field inspection and near-real-time reporting workflows.

Publications & Manuscripts

Hsu, S. H., & Golparvar-Fard, M. (2026). Beyond Cracks: Synthetic Image and Geometry Generation for Computer Vision Detection and Severity Assessment of Diverse Concrete Surface Defects. *Automation in Construction*. Under review.

Hsu, S. H., Jung, Y., Fu, J., & Golparvar-Fard, M. (2025). Self-Evaluation of Single and Multi-Agent LLMs as Assistants in Inspection Reporting Workflows. *Joint CSCE Construction Specialty & CRC Conference*.

*Jung, Y., *Cho, I., ***Hsu, S. H.**, & Golparvar-Fard, M. (2024). VisualSiteDiary: A detector-free Vision-Language Transformer model for captioning photologs. *Automation in Construction*, 165, 105483. (*Equal contribution)

Hsu, S. H., Fu, J., & Golparvar-Fard, M. (2024). VL-Con: Vision-Language Dataset for Deep Learning-based Construction Monitoring Applications. *ISARC*.

Hsu, S. H., & Golparvar-Fard, M. (2024). Requirements for Parametric Design of Physics-Based Synthetic Data Generation for Learning and Inference of Defect Conditions. *Construction Research Congress 2024*.

Hsu, S. H., Hung, H. T., Lin, Y. Q., & Chang, C. M. (2023). Defect inspection of indoor components in buildings using deep learning object detection and augmented reality. *Earthquake Engineering and Engineering Vibration*, 22(1), 41–54.

Hsu, S. H., Chang, T. W., & Chang, C. M. (2022). Impacts of label quality on performance of steel fatigue crack recognition using deep learning-based image segmentation. *Smart Structures and Systems*, 29(1), 207–220.

Technical Skills

Programming: Python, C++17, SQL
Machine Learning: PyTorch, Hugging Face Transformers, scikit-learn, OpenCV, vision-language modeling, LLM/RAG evaluation
Vision / 3D: COLMAP, structure-from-motion, multi-view stereo, 3D reconstruction, synthetic data generation
Systems: CUDA, Linux, Docker, Git, AWS